

The Ferris Wheel Puzzle



Akhila's Rules:

Hoopla rides (y) are exactly half her Giant Wheel rides (x).

Each Wheel ride = ₹3.

Each Hoopla game = ₹4.

Total spent = ₹20.

How do we find the exact number of rides?
We need math language! 🖱️

$$y = \frac{1}{2}x$$

$$3x + 4y = 20$$

Let's crack the code!



The Core Concept



Pair of Linear Equations in Two Variables

Two linear equations that share the exact same two variables (x and y).

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

a, b, and c are just real numbers, and a & b can't both be zero! 🧠

Graphical Solutions: The 3 Possibilities

Two Lines
on a Graph



Intersecting Lines

Exactly **ONE**
unique solution.

(Consistent)



Coincident Lines

INFINITELY many
solutions.

(Dependent/Consistent)



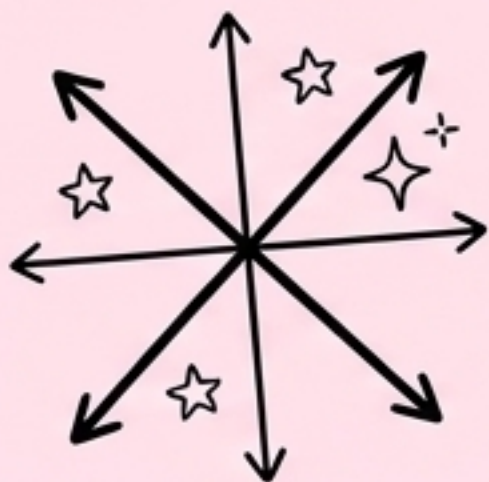
Parallel Lines

NO solution.

(Inconsistent)

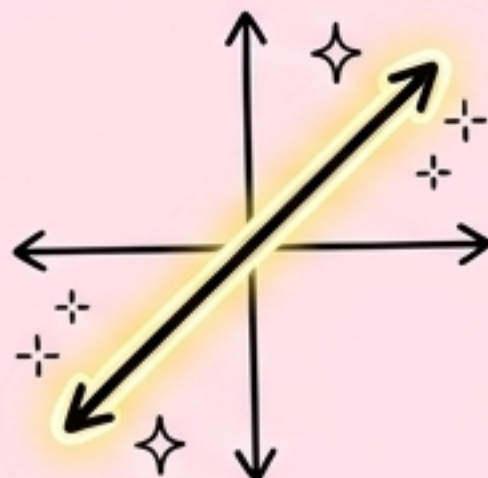
The Ultimate Ratio Cheat Sheet 📌

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$



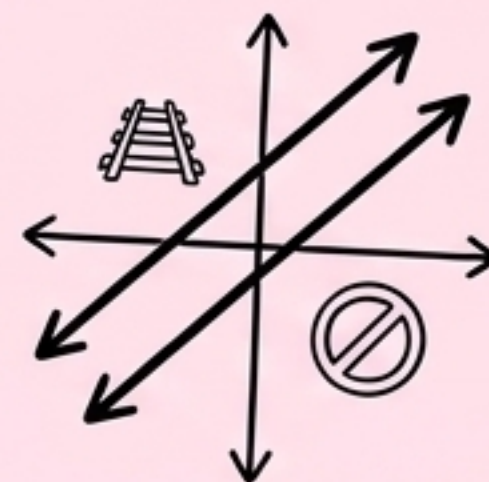
1 Solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$



Infinite Solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$



No Solution

Check the ratios of the coefficients (a, b, c)
to predict the graph instantly! 🚀

Method 1: Substitution

Step 1:
Isolate a
variable.

From $x + 2y = 3 \rightarrow$

$$x = 3 - 2y$$

Step 2:
Substitute into
the other
equation.

$$7(3 - 2y) - 15y = 2$$

Step 3:
Solve, then
plug back in!

$$y = 19/29, x = 49/29$$

Pro-Tip: Always pick the equation where a variable has a coefficient of 1 to isolate first! Saves time! 

Method 2: Elimination



**Step 1:
Multiply to
match
coefficients.**

$\text{Eq1} \times 3, \text{Eq2} \times 4$
to make the y's
match at 12y.



**Step 2:
Add or
Subtract to
Eliminate.**

Poof! The 12y
disappears.
Solve for x.
(x = 2000)

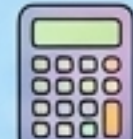
**Step 3:
Solve for the
survivor.**

Plug x = 2000 back
into an original
equation to find y
(y = 4000).

Watch your signs!
If the coefficients have the
same sign, **SUBTRACT**.
Opposite signs? **ADD**.



Word Problem Translator



English:

Math:

Seven years ago...

$$(x - 7) \text{ and } (y - 7)$$

Three times as old...

$$x = 3y$$

Two-digit number...

$$10x + y$$

Fixed charge + per day...

$$x + 3y$$

Save this cheat-sheet for your next exam!

